

A Unified Control Plane for Machine-Learning Security Services

github.com/poojakira/unified-ml-security-platform

Research Brief - Poster 07

Evidence status: Refreshed against current code snapshot
01e58ba257764e47e230a3eecd5a821bb85e7985 and successful CI/CD run 36783840279 on 2026-09-30.

Repository

github.com/poojakira/unified-ml-security-platform - public, default branch main.

Academic Project Title

A Unified Control Plane for Machine-Learning Security Services

Subtitle

An Authenticated Gateway for Independently Released ML Security Products

One-Sentence Contribution

A real authenticated FastAPI control plane with **four production HTTP service slots**, a local contract-test topology built from stubs, and a production Compose contract that requires externally released product images instead of pretending the stubs are product implementations.

Method

1. Authenticate external callers at the gateway.
2. Strip caller-controlled sensitive headers and inject per-service credentials.
3. Route four synchronous service prefixes over an internal-only network.
4. Use local health-contract stubs to test topology without misrepresenting product logic.
5. Require explicit external product images in production Compose.
6. Validate unit, security, integration, image-build, and deployment-contract behavior in CI.

Current Verified Evidence

Current-main Python 3.12 unit-test job reports:

- **67 tests passed.**
- **56.89% statement coverage.**
- Separate local contract-stub health jobs for MCP, LLM red-team, dataset-poison, and model-privacy completed successfully.
- Integration topology tests completed successfully.
- Security scan, dependency audit, production Compose validation, and gateway image build/push job completed successfully.

Critical Claim Correction

The previous poster said the gateway fronted "three real backends." That is not an accurate description of the current repository. The correct boundary is:

- local Compose uses **contract stubs**;
- production Compose defines **four externally supplied real product image slots**;
- stub health checks prove routing/topology contracts only, not downstream product functionality.

Limitations

- Product business logic is owned by external repositories/images.
- No production deployment, uptime, or load SLO is established.
- Multi-replica shared rate limiting/state distribution requires deployment-specific infrastructure.
- Batch tools such as HF provenance scanning and adversarial robustness are intentionally not forced behind synchronous HTTP routes.

Reproducibility

```
git clone https://github.com/poojakira/unified-ml-security-platform.git
cd unified-ml-security-platform

git checkout 01e58ba257764e47e230a3eecd5a821bb85e7985

python -m pip install -e ".[dev]"

pytest tests/ -q --cov=. --cov-report=term

docker compose config

docker compose -f docker-compose.prod.yml config
```

Expected core Python evidence: **67 passed, 56.89% coverage.**